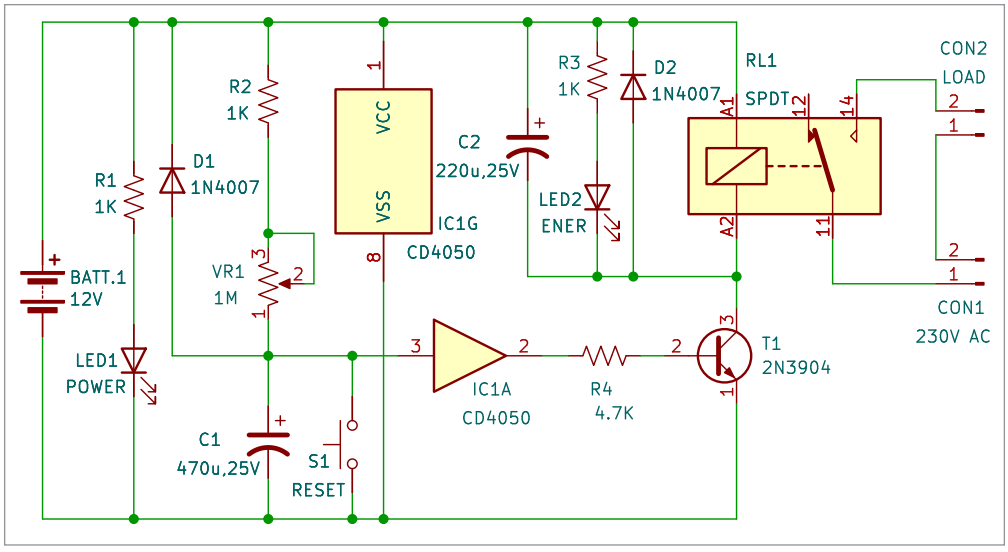


Fig. 3: Circuit diagram of the on-delay timer

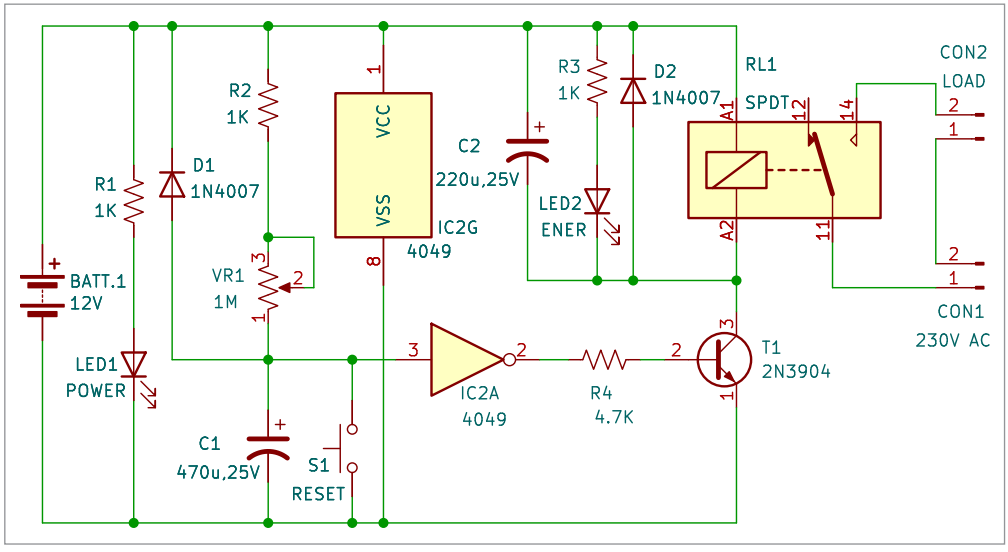


Fig. 4: Circuit diagram of off-delay timer

The delay before the relay energises depends primarily on the time required to charge C1 through R2 and VR1. Increasing either the resistance or capacitance increases the delay, while reducing either value decreases it.

The pinout of the 4049, a hex inverter, is compatible with that of the CD4050 for this application. Therefore, the CD4050 in Fig. 3 can be replaced with a 4049 (IC2) to configure the system as an off-delay timer, as shown in Fig. 4. The IC contains six inverters, but only IC2A is used.

Fig. 4 shows the off-delay timer. It is built around a hex inverter 4049 (IC2). In the off-delay timer configuration, the output of the 4049 is initially high and the relay remains energised. When the voltage across C1 rises above the input threshold of the 4049, its output changes to the low state, switching off transistor T1 and consequently de-energising the relay.

The exact threshold depends on the IC and supply voltage. The delay can be varied by changing the value of C1 or the resistance of R2 and VR1. The charging voltage can also be limited to a fraction of the supply voltage using a voltage divider.

Several additional components are included to ensure reliable operation and provide visual indication. LED1 serves as the power-on indicator, while LED2 indicates that the relay is energised. Capacitor C2 helps prevent relay chatter during switching. Switch S1 is used to provide a discharge path for C1 for quick reuse of the timer.

EFY note. The circuit operates from 5V to 12V. For a 5V supply, use a suitable 5V relay instead of the

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